

Load Balancer

1. Configure Classic Load balancer.

Objective

To configure a Classic Load Balancer and distribute HTTP traffic between multiple EC2 instances.

Purpose

Classic Load Balancer is the previous-generation ELB. It distributes incoming traffic across registered EC2 instances and performs health checks so that traffic can be sent to healthy instances

Steps

Created/selected a VPC.

Created/selected two public subnets in different Availability Zones.

Created two EC2 web servers.

Installed Apache on both EC2 instances.

Created load balancer and EC2 security groups.

Created a Classic Load Balancer.

Configured HTTP port 80 listener.

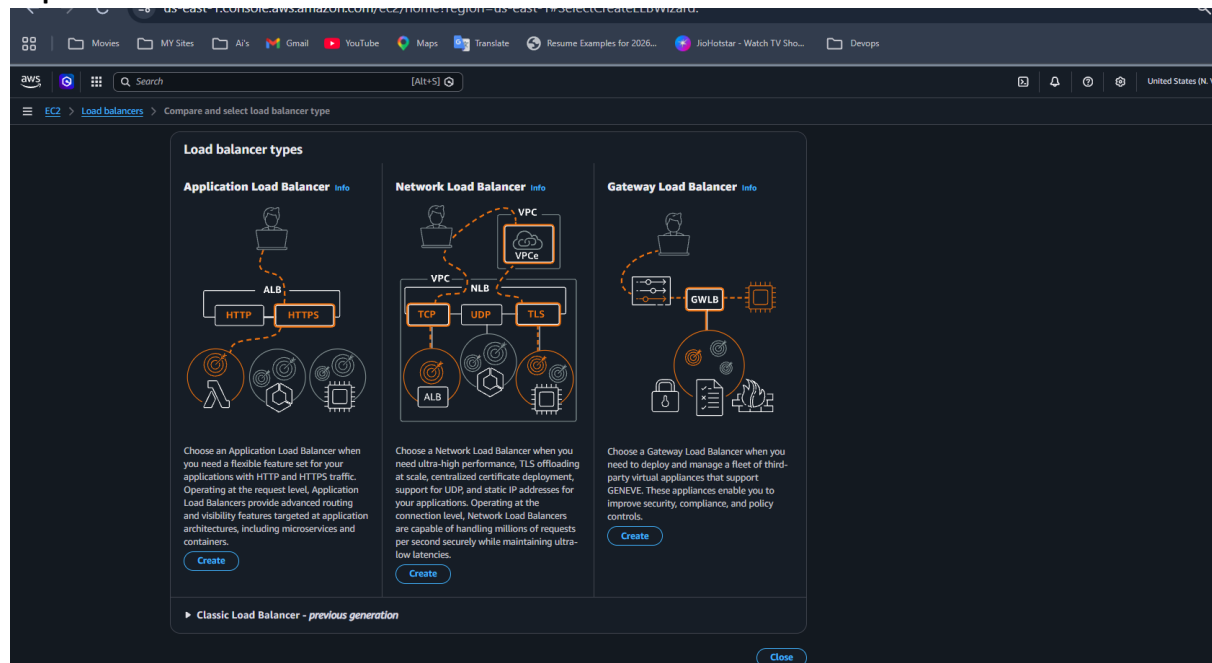
Configured health check.

Registered EC2 instances.

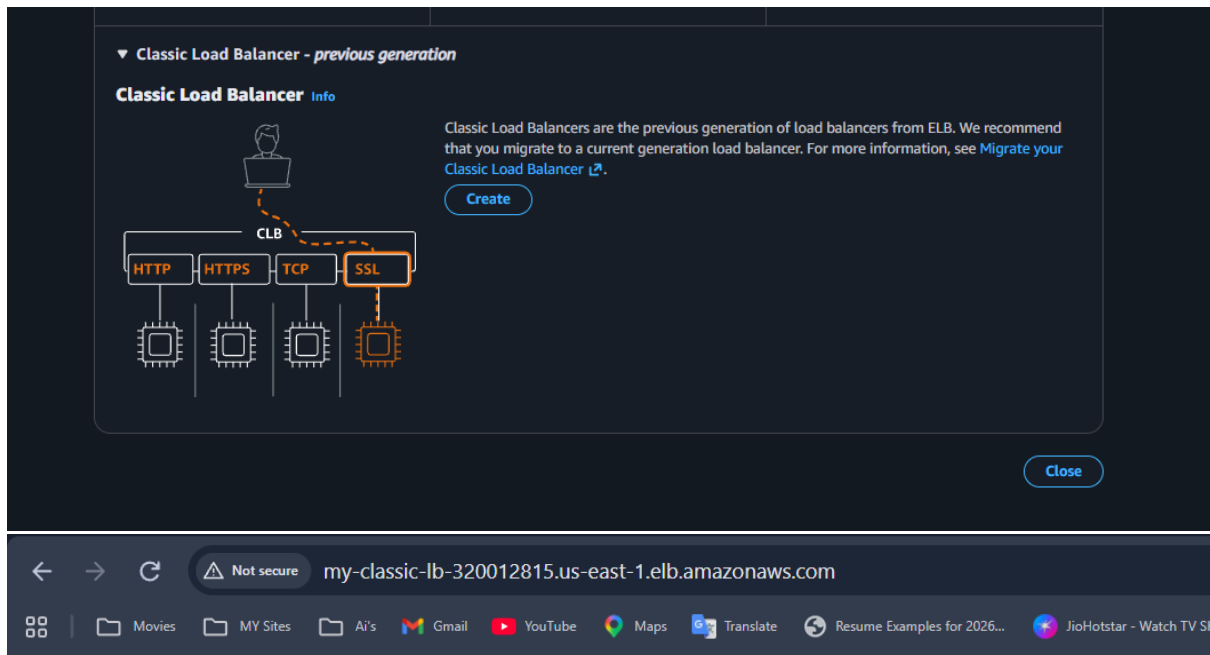
Verified instance health.

Tested the Classic Load Balancer DNS name

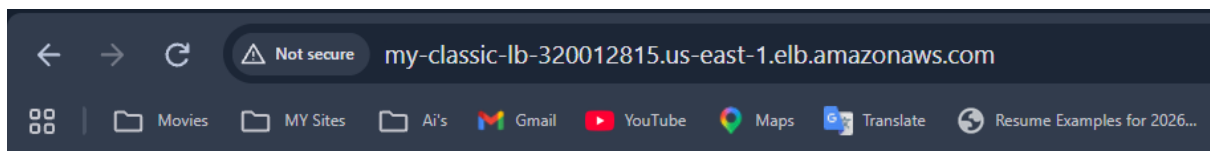
Implementation:



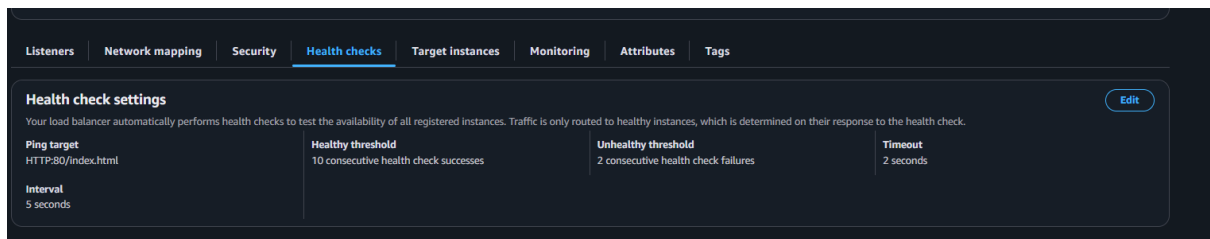
Load Balancer



welcome to server - 2



Welcome to server - 1



Conclusion

The Classic Load Balancer was successfully configured and connected to multiple EC2 instances. The load balancer successfully distributed HTTP requests to healthy registered instances.

2. Configure Application Load balancer.

Objective

To create an Application Load Balancer and route HTTP traffic to multiple EC2 instances using a target group.

Purpose

An Application Load Balancer is designed for HTTP/HTTPS traffic and supports application-level routing features such as host-based and path-based routing

Steps

[Opened EC2 Load Balancers.](#)

[Created an Application Load Balancer.](#)

[Selected Internet-facing scheme.](#)

Load Balancer

Selected IPv4.

Selected two Availability Zones.

Configured load balancer security group.

Created a target group.

Registered two EC2 instances.

Configured HTTP health checks.

Created an HTTP listener on port 80.

Forwarded listener traffic to the target group.

Verified target health.

Tested ALB DNS name.

Implmentation:

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-0c36b75208ae074fd	Server - 2	Running	ssh- httpd -https	us-east-1b	10.0.1.142	subnet-0143f7f5bc7c9f73b
i-00c3ed40ee3fc7788	Server - 1	Running	ssh- httpd -https	us-east-1a	10.0.0.239	subnet-02a3fa30e857e6795

2 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

[Include as pending below](#)

tg-web

Details

arn:aws:elasticloadbalancing:us-east-1:567487919764:targetgroup/tg-web/600badb825b9a4ac

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-0061f3970762db45a
IP address type IPv4	Load balancer None associated		

2	2 Healthy	0 Unhealthy	0 Unused	0 Initial	0 Draining
Total targets					
	0 Anomalous				

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets | Monitoring | Health checks | Attributes | Tags

Registered targets (2) [Info](#)

Anomaly mitigation: Not applicable

[Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

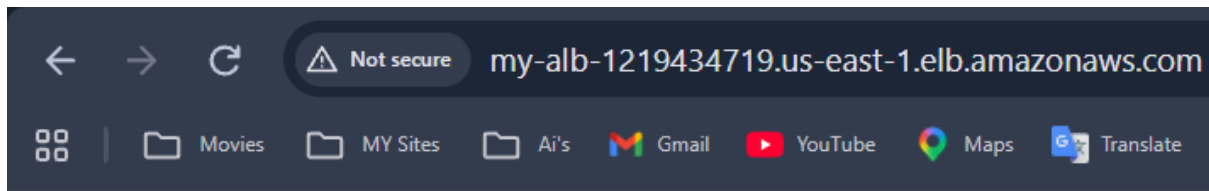
Instance ID	Name	Port	Zone	Health status	Health status details	Administrative o...
i-0c36b75208ae074fd	Server - 2	80	us-east-1b (us...)	Healthy	-	No override
i-00c3ed40ee3fc7788	Server - 1	80	us-east-1a (us...)	Healthy	-	No override

← → ↺ Not secure my-alb-1219434719.us-east-1.elb.amazonaws.com

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Welcome to server - 1

Load Balancer



welcome to server - 2

Conclusion

The Application Load Balancer was successfully configured. Both EC2 instances were registered in the target group and the ALB successfully distributed HTTP requests to healthy targets.

3. Configure Network Load balancer.

Objective

To configure a Network Load Balancer and distribute TCP traffic to multiple EC2 instances.

Purpose

A Network Load Balancer is designed for high-performance Layer 4 traffic such as TCP, UDP and TLS. AWS describes NLBs as the load balancer type for TCP/UDP/TLS workloads where very high performance is required

Steps

Created an NLB target group.

Selected instance targets.

Configured TCP port 80.

Configured health checks.

Registered two EC2 instances.

Created an internet-facing NLB.

Selected two Availability Zones.

Configured TCP listener on port 80.

Forwarded traffic to the target group.

Verified target health.

Tested NLB DNS.

Implementation:

Load Balancer

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

DevOps-NLB

Details

Load balancer type

Network

Scheme

Internet-facing

Status

Active

Hosted zone

Z26RNL4JYFTOTI

VPC

vpc-0f7aebc6164408785

Load balancer IP address type

IPv4

Availability Zones

subnet-015c3f9a9cb76715c us-east-1b (use1-az2)

subnet-0ad12f365397085ba us-east-1a (use1-az1)

DNS name

DevOps-NLB-34a4a86c991c4e28.elb.us-east-1.amazonaws.com (A Record)

Load balancer ARN

arn:aws:elasticloadbalancing:us-east-1:520426835611:loadbalancer/net/DevOps-NLB/34a4a86c991c4e28

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1)

A listener checks for connection requests using the protocol and port that you configure. Traffic received by a Network Load Balancer listener is forwarded to the selected target group.

Filter listeners

Protocol/Port

Default action

Rules

ARN

Security policy

Default SSL/TLS certificate

Tags

TCP:80

Forward to target group

DevOps-TG (1: 100%)

Target group stickiness: Off

1 rule

ARN

Not applicable

Not applicable

0 tags

Network mapping

Targets in the listed zones and subnets are available for traffic from the load balancer using the IP addresses shown.

VPC

vpc-0f7aebc6164408785

IPv4 VPC CIDR: 10.0.0.0/16

IPv6 VPC CIDR: -

Load balancer IP address type

IPv4

Availability Zones and subnets

Including two or more Availability Zones, and corresponding subnets, increases the fault tolerance of your applications.

Zone

Subnet

IPv4 address

Private IPv4 address

IPv6 address

us-east-1b (use1-az2)

subnet-015c3f9a9cb76715c

Assigned by AWS

Assigned from CIDR 10.0.11.0/...

Not applicable

us-east-1a (use1-az1)

subnet-0ad12f365397085ba

Assigned by AWS

Assigned from CIDR 10.0.1.0/24

Not applicable

Resource map

View, explore, and troubleshoot your load balancer's architecture.

Overview

Unhealthy target map

Show resource details

DevOps-NLB

Listeners (1)

TCP:80

Target groups (1)

Instance, TCP

DevOps-TG

2 targets

Targets (2)

i-057b0769c2306883b

Port 80

Healthy

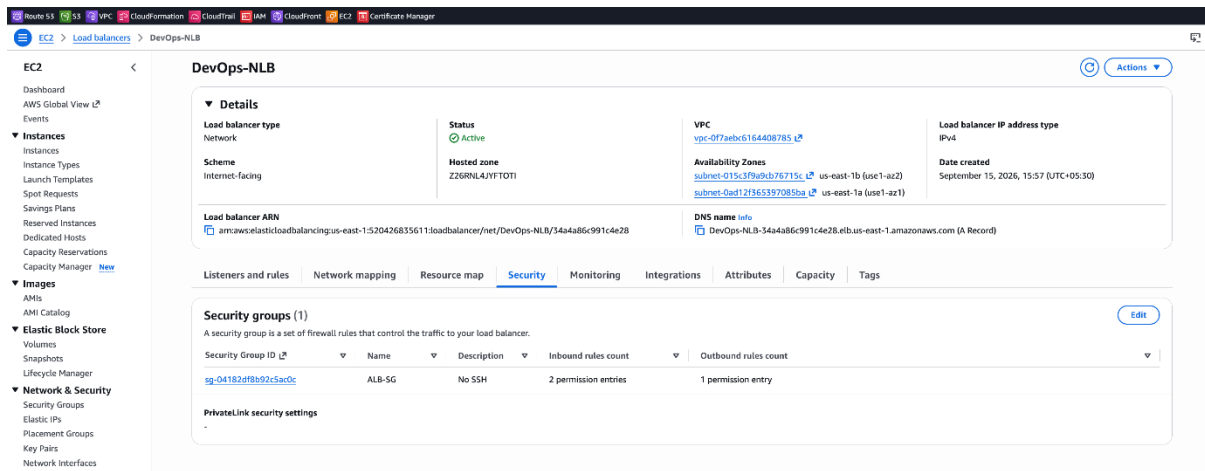
i-08cb48badc7a3bdd6

Port 80

Healthy

Last fetched seconds ago [Export](#)

Load Balancer



Conclusion

The Network Load Balancer was successfully configured. TCP traffic received on port 80 was forwarded to healthy EC2 instances through the NLB target group.

4. Attach SSL for application load balancer.

Objective

To configure HTTPS on the Application Load Balancer using an SSL/TLS certificate from AWS Certificate Manager.

Purpose

SSL/TLS encrypts communication between clients and the Application Load Balancer. The ALB can terminate HTTPS and forward traffic to EC2 over HTTP or HTTPS depending on the architecture.

Steps

Opened AWS Certificate Manager.

Requested a public certificate.

Entered the domain name.

Selected DNS validation.

Created the ACM validation record in Route 53.

Waited for the certificate to become issued.

Opened the Application Load Balancer.

Added an HTTPS listener.

Configured port 443.

Selected the ACM certificate.

Forwarded HTTPS traffic to the web target group.

Allowed HTTPS port 443 in the ALB security group.

Load Balancer

Secure listener settings

Security policy

Your load balancer uses a Secure Socket Layer (SSL) negotiation configuration called a security policy to manage SSL connections with clients. [Compare security policies](#)

Security category

All security policies

Policy name

ELBSecurityPolicy-TLS13-1-2-Res-PQ-2025-09 (Recommended)

Default SSL/TLS server certificate

The certificate used if a client connects without SNI protocol, or if there are no matching certificates. You can source this certificate from AWS Certificate Manager (ACM), Amazon Identity and Access Management (IAM), or import a certificate. This certificate will automatically be added to your listener certificate list.

Certificate source

From ACM

From IAM

Import certificate

Certificate (from ACM)

The selected certificate will be applied as the default SSL/TLS server certificate for this load balancer's secure listeners.

sunith.online

b637f95b-b52d-4bae-b9d7-3c8e0dbd3fa7

Request new ACM certificate

Client certificate handling

Client certificates are used to make authenticated requests to remote servers. [Learn more](#)

Mutual authentication (mTLS)

Mutual TLS (Transport Layer Security) authentication offers two-way peer authentication. It adds a layer of security over TLS and allows your services to verify the client that's making the connection.

Passthrough

The entire certificate is sent to the backend target as HTTP headers to verify whether the client certificate is trustworthy.

Verify with trust store

The load balancer and client verify each other's identity and establish a TLS connection to encrypt communication between them.

Choose this method if you are confident in the process of creating records and know which options you need.

Choose this method if you need more explanations as you create your record.

Create record

Quick create record

Record 1

Record name

subdomain

sunith.online

Record type

A - Routes traffic to an IPv4 address and some AWS resources

Keep blank to create a record for the root domain.

Alias

Route traffic to

Alias to Application and Classic Load Balancer

US East (N. Virginia)

Q dualstack.my-alb-1219434719.us-east-1.elb.amazonaws.com

Use: "dualstack.my-alb-1219434719.us-east-1.elb.amazonaws.com"

dualstack.my-alb-1219434719.us-east-1.elb.amazonaws.com

dualstack.my-classic-lb-320012815.us-east-1.elb.amazonaws.com

Add another record

Cancel

Create records

← → ↺

sunith.online

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📁 MY Sites

📁 Ai's

📧 Gmail

📺 YouTube

📍 Maps

🗨️ Translate

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🌈

Welcome to server - 1

Route 53

Dashboard

Hosted zones

Health checks

Profiles

Global Resolver

VPC Resolver

Domains

IP-based routing

Traffic flow

DNS Firewall

Application Recovery Controller

sunith.online

Delete zone

Test record

Configure query logging

Hosted zone details

Edit hosted zone

Records (4)

Accelerated recovery

DNSSEC signing

Hosted zone tags (0)

Records (4)

Filter records by property or value

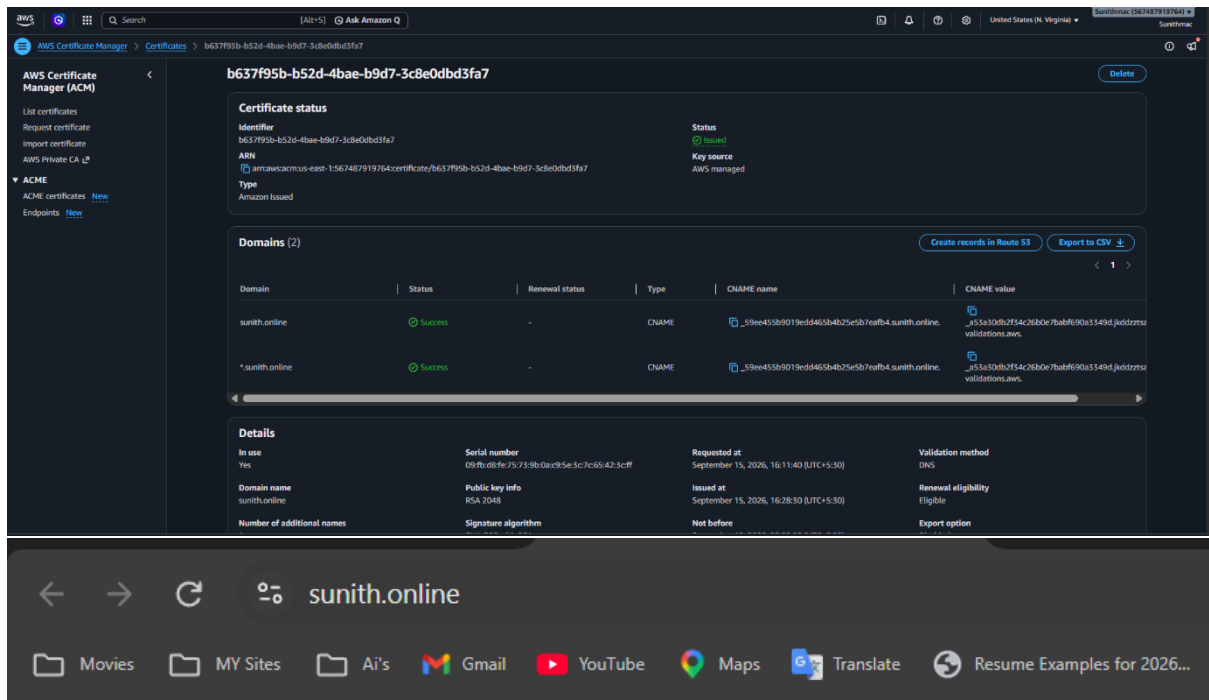
Type

Routing p...

Alias

Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...	Evaluat...	Recor...
sunith.online	A	Simple	-	Yes	dualstack.my-alb-12194347...	-	-	Yes	-
sunith.online	NS	Simple	-	No	rs-850.amazonaws-42.net. rs-32.amazonaws-04.com. rs-1041.amazonaws-02.org. rs-1878.amazonaws-42.co.uk.	172,800	-	-	-
sunith.online	SOA	Simple	-	No	rs-850.amazonaws-42.net. awsd...	900	-	-	-
_59ee455b9019ed0465b4b25e5b7eafbf4.sunith.online	CNAME	Simple	-	No	_a53a50db7f54c2680e7baaf...	300	-	-	-

Load Balancer



welcome to server - 2

Conclusion

SSL/TLS was successfully configured for the Application Load Balancer using AWS Certificate Manager. The ALB can now accept secure HTTPS connections on port 443

5. Map Application load balancer to R53.

Objective

To connect a custom domain name to the Application Load Balancer using Amazon Route 53.

Purpose

The purpose is to provide users with a human-readable domain name instead of the AWS-generated load balancer DNS name.

Steps

Opened Route 53.

Opened the hosted zone.

Selected the required domain.

Created an A record.

Enabled Alias.

Selected Application Load Balancer.

Selected the correct AWS region.

Selected the ALB.

Configured target health evaluation.

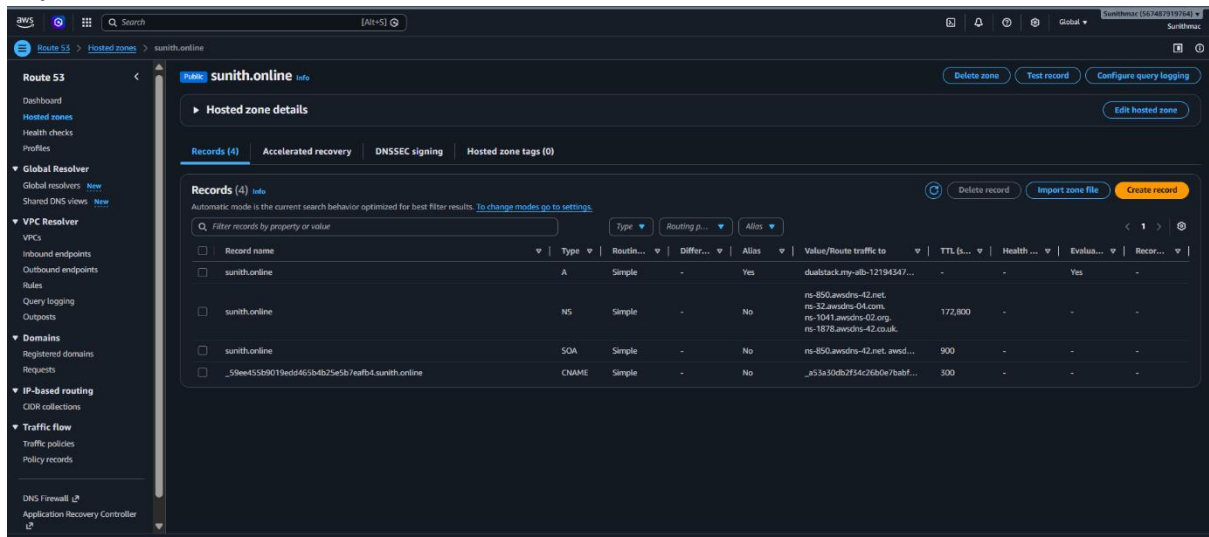
Created the record.

Load Balancer

Waited for DNS propagation.

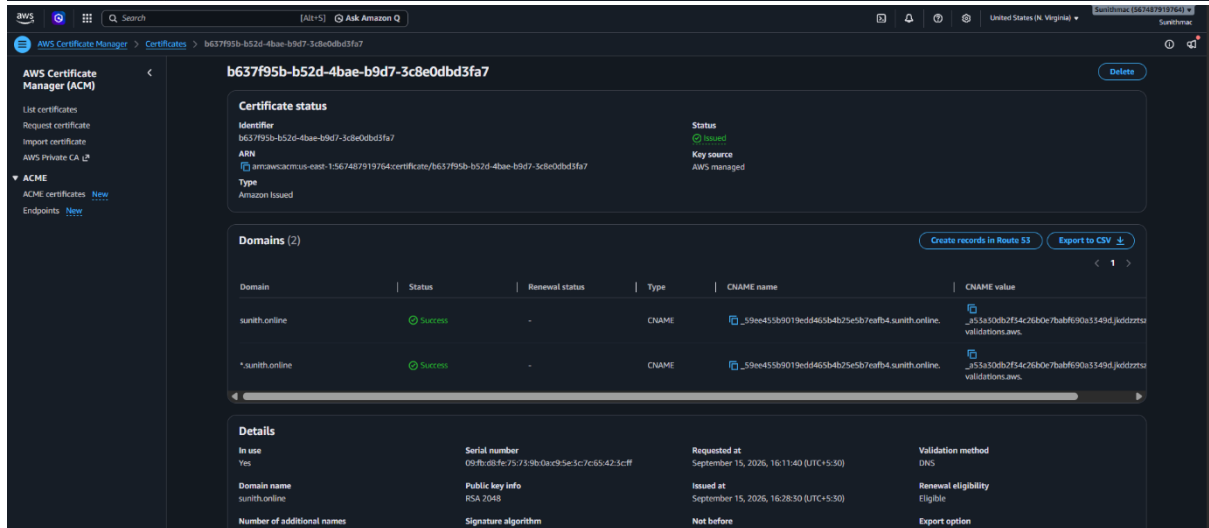
Tested the custom domain.

Implementation:

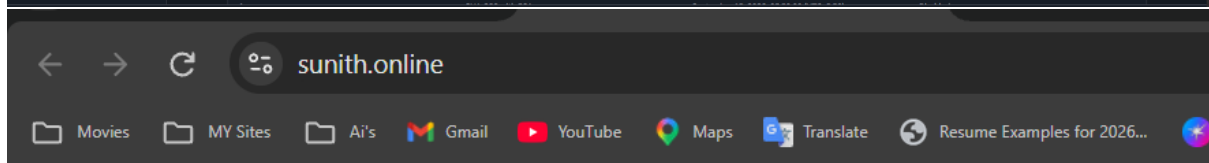


The screenshot shows the AWS Route 53 console for the hosted zone `sunith.online`. The "Records (4)" tab is selected, displaying a table of DNS records:

Record name	Type	Routing	Differ...	Alias	Value/Route traffic to	TTL (s...)	Health ...	Evalu...	Recor...
<code>sunith.online</code>	A	Simple	-	Yes	<code>duabstack.my-ali-12194347...</code>	-	-	-	-
<code>sunith.online</code>	NS	Simple	-	No	<code>rs-850.awsdns-42.net</code> <code>rs-32.awsdns-04.com</code> <code>rs-1041.awsdns-02.org</code> <code>rs-1878.awsdns-42.co.uk</code>	172,800	-	-	-
<code>sunith.online</code>	SOA	Simple	-	No	<code>rs-850.awsdns-42.net. awsd...</code>	900	-	-	-
<code>_59ee453b9019edd465b4b25e5b7eafb4.sunith.online</code>	CNAME	Simple	-	No	<code>_a53a30db2f54c26b0e7ba6f...</code>	300	-	-	-

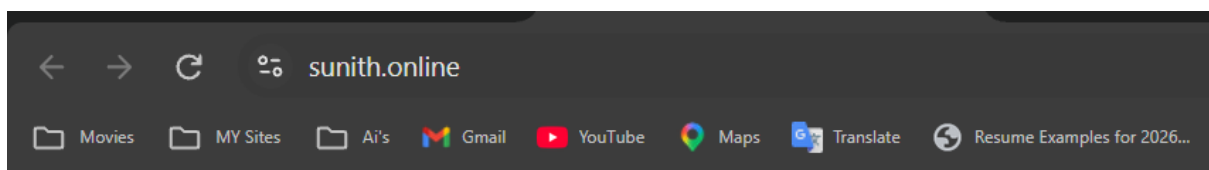


The screenshot shows the AWS Certificate Manager console for the certificate `b637f95b-b52d-4bae-b9d7-3c8e0dbd3fa7`. The "Certificate status" section shows the certificate is "Issued". The "Domains (2)" section lists the domains `sunith.online` and `*.sunith.online`, both with a status of "Success". The "Details" section provides information about the certificate, including the serial number, public key info, and validation method.



The screenshot shows a web browser with the address bar displaying `sunith.online`. The browser's taskbar at the bottom shows various application icons, including Movies, MY Sites, Ai's, Gmail, YouTube, Maps, Translate, and Resume Examples for 2026...

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The screenshot shows a web browser with the address bar displaying `sunith.online`. The browser's taskbar at the bottom shows various application icons, including Movies, MY Sites, Ai's, Gmail, YouTube, Maps, Translate, and Resume Examples for 2026...

welcome to server - 2

Conclusion

Load Balancer

The Application Load Balancer was successfully mapped to the custom Route 53 domain using an Alias A record. Users can now access the application using the custom domain name

6. Push the application load balancer logs to S3.

Objective

To enable Application Load Balancer access logging and store the logs in an Amazon S3 bucket.

Purpose

The purpose is to maintain a centralized location for ALB request logs that can be used for troubleshooting, monitoring, auditing, and traffic analysis.

Steps

Created an S3 bucket.

Selected the same region as the ALB.

Enabled private access.

Configured SSE-S3 encryption.

Created the required S3 bucket policy.

Granted the ELB log delivery service permission to write logs.

Opened the Application Load Balancer.

Opened the Attributes section.

Enabled Access Logs.

Specified the S3 bucket URI.

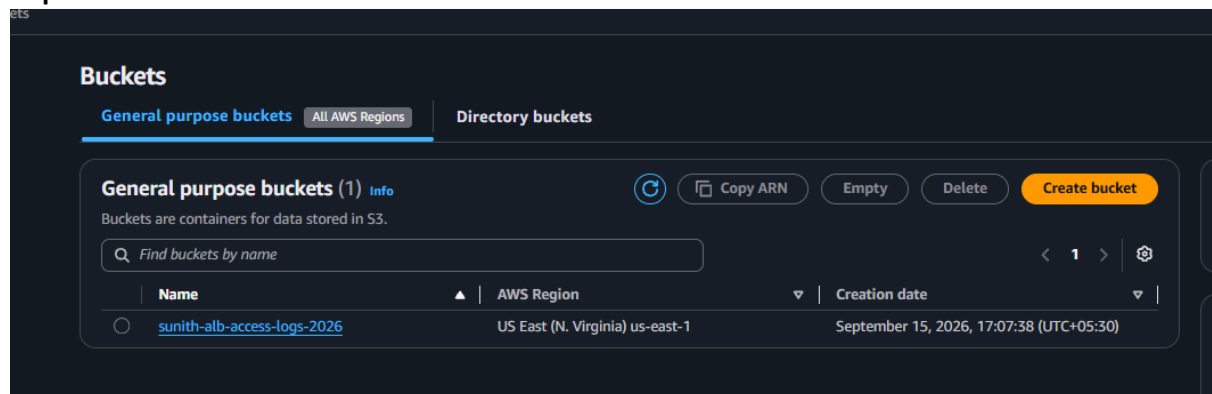
Saved the ALB configuration.

Generated application traffic.

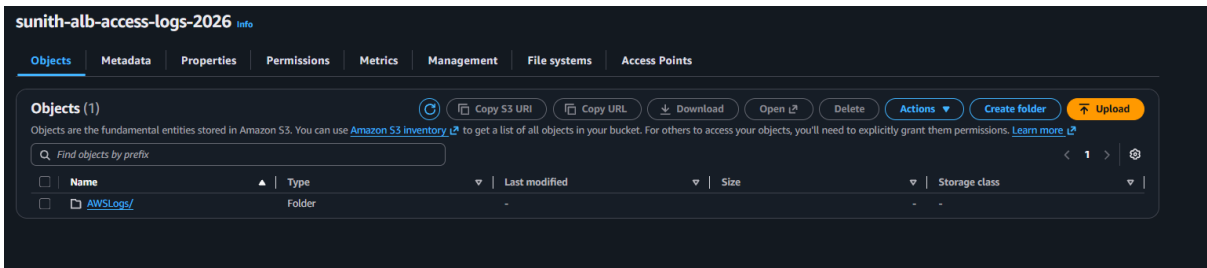
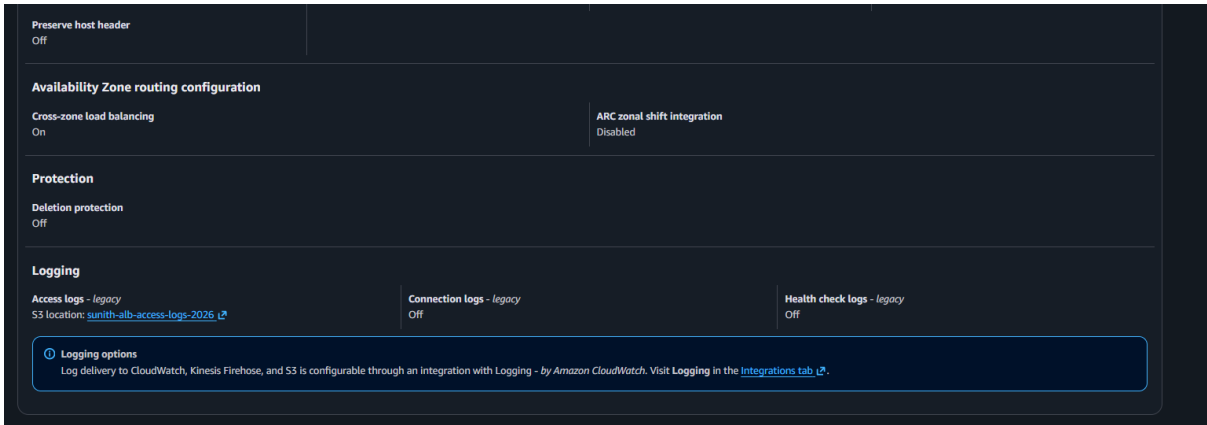
Opened the S3 bucket.

Verified ALB log files.

Implementation:



Load Balancer



Conclusion

Application Load Balancer access logging was successfully configured. ALB request logs are now delivered to the S3 bucket for monitoring, troubleshooting, auditing, and analysis.